Yenwanth Reddy Anumula

+1 (704)-241-9937|yenwanthreddy359@gmail.com| Linkedin

EDUCATION

University of North Carolina Charlotte May2027
Master of Science in Computer Science
VNR Vignana Jyothi Institute of Engineering and Technology Aug 2025
Bachelor of Technology in Computer Science
Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Cloud Computing, Machine Learning, Deep Learning, Statistics, Computer Vision

INTERNSHIPS

AWS Cloud Intern Mar2024 – Jun2024
F13 Technologies

- Designed scalable AWS architectures using EC2 and IAM following cloud optimization best practices.
- Improved infrastructure performance by 20% through resource provisioning and monitoring.
- Developed and deployed a recommendation system using Amazon Personalize, increasing user engagement by 30%.
- Implemented ML model deployment pipelines on cloud infrastructure.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++, Java, JavaScript, SQL
Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, PyTorch, CNN, 3D U-Net, VGG16, Transfer Learning, NLP, Data Augmentation, Model Evaluation
Computer Vision & NLP: OpenCV, Image Segmentation, Object Detection, Text Classification, Information Extraction
Data Science & Algorithms: Statistical Modeling, Feature Engineering, Optimization Algorithms, Supervised & Unsupervised Learning, Algorithm Design, Time & Space Complexity Analysis
Cloud & Systems: AWS EC2, IAM, Amazon Personalize, Scalable Architecture Design, Linux, Distributed Systems Fundamentals

PUBLICATIONS & RESEARCH

Early Identification of Critical Health Conditions of Breast, Chest, and Brain through Medical Imaging
Conference: ICAML 2025 (Undergoing Publication)
Hand Gesture Video Controller, presented in: Springer ACSAR Proceedings

PROJECTS

AI-Based Medical Imaging Research (Computer Vision | Deep Learning)

- Developed a 3D U-Net CNN model for brain tumor and lung nodule segmentation, improving diagnostic precision and reducing processing time by 30%.
- Applied transfer learning (VGG16) to enhance classification accuracy by 25% across breast, chest, and brain imaging datasets.
- Implemented large-scale data augmentation pipelines to improve model generalization across diverse medical datasets.
- Designed a real-time inference system optimized for low latency and scalable deployment.
- Conducted experimental evaluations using performance metrics including precision, recall, F1-score, and ROC-AUC.

NLP-Based Medical Diagnosis System (Natural Language Understanding)

- Built an NLP pipeline to extract structured medical insights from unstructured clinical text.
- Applied supervised ML models to classify disease conditions with 90% accuracy.
- Designed information extraction workflows improving data processing efficiency by 21%.
- Conducted comparative model evaluation across multiple algorithms.

Real-Time Hand Gesture Recognition (Computer Vision | HCI)

- Designed a CNN-based real-time gesture recognition system using OpenCV (98% accuracy).
- Engineered low-latency inference pipeline for interactive systems.
- Addressed robustness challenges (lighting, background noise, edge cases).
- Optimized algorithmic performance for real-time scalability.

CERTIFICATIONS

- AWS Cloud Practitioner Essentials
- Python 101 for Data Science
- Software Engineering by Goldman Sachs